

# Zitao Chen

zitaoc@ece.ubc.ca | <https://zitaoc.github.io>

## PROFESSIONAL EXPERIENCE

---

|                                                               |                     |
|---------------------------------------------------------------|---------------------|
| Assistant Professor, Department of EECS, University of Kansas | Oct 2025 - present  |
| Research Technician, University of British Columbia           | Jul 2020 - Feb 2021 |

## EDUCATION

---

### University of British Columbia

|                                                |                     |
|------------------------------------------------|---------------------|
| Ph.D. in Electrical and Computer Engineering   | Jan 2022 - Oct 2025 |
| M.A.Sc. in Electrical and Computer Engineering | Sep 2018 - Jun 2020 |
| Advisor: <a href="#">Karthik Pattabiraman</a>  |                     |

### China University of Geosciences (Wuhan)

|                                |                     |
|--------------------------------|---------------------|
| B.Eng. in Information Security | Sep 2014 - Jun 2018 |
|--------------------------------|---------------------|

## RESEARCH INTERESTS

---

Trustworthy machine learning; Cybersecurity

## PUBLICATIONS [[Google Scholar](#)]

---

- [CCS'25] Zitao Chen, Karthik Pattabiraman "Anonymity Unveiled: A Practical Framework for Auditing Data Use in Deep Learning Models" *In Proceedings of the 2025 ACM Conference on Computer and Communications Security*. Acceptance rate: 13.9% [Paper] [Code]  
[Earned all Artifact Badges: Artifact Available, Functional and Results Reproduced](#)
- [NDSS'25] Zitao Chen, Karthik Pattabiraman "A Method to Facilitate Membership Inference Attacks in Deep Learning Models" *The ISOC Network and Distributed Systems Security Symposium, 2025*. Acceptance rate: 16.1% [Paper] [Code]  
[Earned all Artifact Badges: Artifact Available, Functional and Results Reproduced](#)
- [NDSS'24] Zitao Chen, Karthik Pattabiraman "Overconfidence is a Dangerous Thing: Mitigating Membership Inference Attacks by Enforcing Less Confident Prediction" *The ISOC Network and Distributed Systems Security Symposium, 2024*. Acceptance rate: 15% [Paper] [Code]  
[Earned all Artifact Badges: Artifact Available, Functional and Results Reproduced](#)
- [AsiaCCS'23] Zitao Chen, Pritam Dash, Karthik Pattabiraman "Jujutsu: A Two-stage Defense against Adversarial Patch Attacks on Deep Neural Networks" *The 18th ACM ASIA Conference on Computer and Communications Security, 2023*. Acceptance rate: 16% [Paper] [Code]
- [DSN'21] Zitao Chen, Guanpeng Li, Karthik Pattabiraman "A Low-cost Fault Corrector for Deep Neural Networks through Range Restriction" *The 51st Annual IEEE/IFIP International Conference on Dependable Systems and Networks, 2021*. Acceptance rate: 16.3% [Paper] [Code]  
[Best Paper Award Runner-up \(2 out of 295 submissions\)](#)  
[Selected for IEEE Top Picks in Test and Reliability \(1 of 7 papers\)](#)  
[Invited for submission to the IEEE Design & Test \(DnT\) journal](#)  
[Our algorithm \(called Ranger\) was adopted by Intel's OpenVINO \[Details\]](#)

- [DSN'21] Pritam Dash, Guanpeng Li, **Zitao Chen**, Mehdi Karimi, Karthik Pattabiraman **"PID-Piper: A Framework for Recovering Robotic Vehicles From Physical Attacks"** *The 51st Annual IEEE/IFIP International Conference on Dependable Systems and Networks*, 2021. Acceptance rate: 16.3% [Paper] [Code]  
Best Paper Award (1 out of 295 submissions)
- [ISSRE'20] **Zitao Chen\***, Niranjhana Narayanan\*, Bo Fang, Guanpeng Li, Karthik Pattabiraman, Nathan DeBardeleben, **"TensorFI: A Flexible Fault Injection Framework for TensorFlow Applications"** *The 31st IEEE International Symposium on Software Reliability Engineering*, 2020. Acceptance rate: 25.7% [Paper] [Code]
- [SC'19] **Zitao Chen**, Guanpeng Li, Karthik Pattabiraman, Nathan DeBardeleben **"BinFI: An Efficient Fault Injector for Safety-Critical Machine Learning Systems"** *In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*, 2019. Acceptance rate: 20.9% [Paper] [Code]  
Finalist for SC Reproducibility Initiative (3 out of 344 submissions)

#### Journal papers-----

- [TDSC] Pritam Dash, Guanpeng Li, **Zitao Chen**, Mehdi Karimi, Karthik Pattabiraman **"Feed-Forward Controller-Based Recovery for Robotic Vehicles from Physical Attacks"** *IEEE Transactions on Dependable and Secure Computing*, 2025
- [DnT] **Zitao Chen**, Guanpeng Li, Karthik Pattabiraman **"A Low-cost Fault Corrector for Deep Neural Networks through Range Restriction"** *IEEE Design & Test*, 2025 (Invited submission to IEEE DnT based on our DSN'21 paper)
- [TDSC] Niranjhana Narayanan, **Zitao Chen**, Bo Fang, Guanpeng Li, Karthik Pattabiraman, and Nathan DeBardeleben **"Fault Injection for TensorFlow Applications"** *IEEE Transactions on Dependable and Secure Computing*, 2022 [Code]
- [FGCS] **Zitao Chen**, Wei Ren, Yi Ren and Kim-Kwang Raymond Choo **"LiReK: A Lightweight and Real-time Key Establishment Scheme for Wearable Embedded Devices by Gestures or Motions"** *Future Generation Computer Systems*, 2018

#### Short papers -----

- [CCS'24 Doctoral Symposium] **Zitao Chen** **"Catch Me if You Can: Detecting Unauthorized Data Use in Training Deep Learning Models"** *In Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS'24)*. 3 pages. 2024
- [IOLTS'20] Karthik Pattabiraman, Guanpeng Li, **Zitao Chen**, **"Error Resilient Machine Learning for Safety-Critical Systems: Position Paper"** *IEEE 26th International Symposium on On-Line Testing and Robust System Design*, 4 pages, 2020. Invited paper

#### HONORS AND AWARDS

- |                                                                                                        |      |
|--------------------------------------------------------------------------------------------------------|------|
| • UBC Applied Science Dean's Graduate Thesis Award                                                     | 2026 |
| ◦ Awarded to the best PhD thesis among the 11 schools & departments in UBC Applied Science.            |      |
| • ACM CCS Young Scholars Development Travel Grant                                                      | 2025 |
| • ACM CCS Student Travel Grant                                                                         | 2025 |
| • IEEE Top Picks in Test and Reliability (1 of 7 papers)                                               | 2024 |
| ◦ Recognizing the most impactful publications in Computer Systems Reliability from 2018-2024           |      |
| ◦ Selected in the final list of Top Picks and invited for submission to the IEEE Design & Test journal |      |
| • DAAD AInet Fellowship (50 awardees worldwide)                                                        | 2024 |

- **Brandwajn Graduate Fellowship (twice)** (given to the top-ranked student in the ECE dept) 2023, 2024
- **ACM CCS Doctoral Symposium Travel Grant** 2024
- **UBC Public Scholar Award** (45 awardees university-wide) 2022
- **UBC Four Year Doctoral Fellowship** (awarded to the top PhD applicants) 2022
- **Best Paper Award at DSN** (DSN is the flagship venue in Dependable Computing) 2021
- **Best Paper Award Runner-up at DSN** (DSN is the flagship venue in Dependable Computing) 2021
- **Finalist for SC Reproducibility Initiative** (SC is the flagship venue in High Perf. Computing) 2019
- **UBC Faculty of Applied Science Graduate Award** 2019-2024

## TEACHING EXPERIENCE

|                           |                                                                   |      |
|---------------------------|-------------------------------------------------------------------|------|
| <b>Instructor</b>         | Trustworthy Machine Learning (University of Kansas)               | 2026 |
| <b>Instructor</b>         | Introduction to Artificial Intelligence (University of Kansas)    | 2026 |
| <b>Teaching assistant</b> | Building Modern Web Applications (University of British Columbia) | 2019 |
| <b>Guest lecturer</b>     | Adversarial Machine Learning (Texas State University)             | 2024 |

## INVITED TALKS

|                                            |                              |              |
|--------------------------------------------|------------------------------|--------------|
| Technical University of Berlin, Germany    | Host: Prof. Konrad Rieck     | October 2024 |
| Technical University of Darmstadt, Germany | Host: Prof. Thomas Schneider | October 2024 |

## SERVICES

### Conference organization

- Co-chair of the Artificial Evaluation Committee @ DSN'27 2027
- Co-chair of the IEEE Workshop on Dependable and Secure Machine Learning @ DSN'26 2026

### Program committee

- USENIX Security Symposium 2027
- The ISOC Network and Distributed System Security Symposium (NDSS) 2026, 2027
- ACM Conference on Computer and Communications Security (CCS) 2026
- The Annual IEEE/IFIP Intl' Conference on Dependable Systems and Networks (DSN) 2026, 2027
- The IEEE European Symposium on Security and Privacy (EuroS&P) 2026
- IEEE International Symposium on Software Reliability Engineering (ISSRE) 2026
- ACM/SIGAPP Symposium On Applied Computing (SAC) 2026, 2027
- The European Dependable Computing Conference (EDCC) 2026
- ACM Workshop on Large AI Systems & Models with Privacy and Security Analysis @ CCS 2025, 2026
- IEEE Workshop on Reliable and Secure AI for Software Engineering @ ISSRE 2025

### Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC) 2025, 2026
- IEEE Transactions on Information Forensics & Security (TIFS) 2026
- IEEE Transactions on Computers (TC) 2026
- IEEE Transactions on Parallel and Distributed Systems (TPDS) 2025
- IEEE Trans on Computer-Aided Design of Integrated Circuits and Systems (TCAD) 2023, 2024, 2026
- ACM Transactions on Intelligent Systems and Technology 2026
- Elsevier Neural Networks 2025
- Elsevier Computer & Security 2024, 2025